

Siddharth Saha

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EDUCATION

Northwestern University, McCormick School of Engineering, Evanston, IL
BS in Computer Science | Mathematics Minor

Graduated June **2025**

GPA: **3.89/4.00**

SKILLS

Languages: Python, TypeScript, JavaScript, C, C++, Java, SQL, C#, MATLAB

AI / Agent Frameworks: LangChain, LangGraph, OpenAI API, PyTorch, Scikit-Learn, NumPy, Pandas, HuggingFace

Agentic Patterns, Models: RAG, tool-use / function calling, multi-step orchestration, LSTMs, RNNs, CNNs, Diffusion Models

Cloud & Backend: AWS (EC2, Lambda, S3), GCP, Docker, MongoDB, Firebase, Google Colab, React, Node.js, Flask, Streamlit

WORK EXPERIENCE

WallachBeth Capital, *Equity Derivatives Trader*

July 2025 - Present

- Build Python automation pipelines to parse, validate, and route trade confirmations across internal and external systems — reducing manual processing time and operational errors
- Facilitate large institutional options trades, employing algorithmic trading to achieve best execution for customers
- Develop scripts to generate bespoke customer reports that provide insight into sector-specific option pricing trends

Xenon Health, *ML Software Engineering Intern*

June 2024 - Sep 2024

- Developed **medical billing platform** using REACT to streamline revenue logging for the use of **100+ employees**
- Utilized S3 bucket for file-storage and AWS lambda functions to perform NLP analysis on healthcare documents
- Developed PyTorch-based LSTM model to analyze revenue trends and detect unprofitable company sites

AI4School, *Software Engineering Intern*

June 2023 - Sep 2023

Ludwig Maximilian Institute - Munich, Germany

- Engineered web-learning platform for edtech **startup** teaching **500+ students** interactive physics
- Contributed to extensive Javascript codebase to calculate student performance metrics to be stored in user database
- Developed RNN model on pytorch to dynamically track student performance and provide personalized feedback

IoT Leaks Research Project, *Security and Privacy Intern*

June 2022 – Sep 2022

Computer Science Department, Northwestern University

- Investigated **privacy** threats to **1000 IoT systems**, publicly accessible via the *shodan.io* API
- Created framework to connect to each IoT system via MQTT protocol, collect data packets passing through devices
- Conducted sensitivity analysis on collected data with TF-IDF vectorization and NLP models (LDA and NMF)

TECHNICAL PROJECTS

Domain-Specific Research Chatbot, *Web Developer*

Jan 2024-March 2024

- Implemented a chatbot based on Retrieval Augmented Generation to cater to journalistic research
- Built a framework to scrape 1000 user-defined articles from trusted sources using the Google News API
- Integrated Chroma VectorDB and OpenAI API for the backend; used Streamlit framework for the frontend

App for Diagnosing Skin Cancer, *Lead Software Developer*

Dec 2023-June 2024

- In team of 7, created **telehealth app** capable of diagnosing 8 categories of skin cancers
- Trained a Convolutional Neural Network using ResNet-18 architecture on **healthcare datasets** to classify skin lesions
- Developed a mobile interface with REACT and a backend using Flask, integrating Firebase for image storage

Additional Projects, *Software and Machine Learning*

Sep 2021- Present

- **Tenant Aid Chatbot**: LLM-powered assistant with ChatGPT API/web-scraping to route tenants to housing resources
- **Diffusion Generative Model**: trained from scratch to produce scenic landscape images
- **Custom Language Interpreter**: built from scratch with recursion, garbage collection, and first-class functions
- **Food Rating Platform**: developed backend for 5,000 Northwestern students with MongoDB and Selenium scraper

LEADERSHIP AND AWARDS

Most Impactful Software Product Award, Northwestern IEEE Technical Showcase

June 2024

Technical Director, Institute of Electrical and Electronics Engineers (IEEE) Executive Board

June 2023-Present

Memberships: Responsible AI Club, Software Dev for Social Change, Northwestern Quant Tech Club

Sep 21- Present

Best Design Award, Northwestern School of Engineering Projects

Dec 2021